

Mathematics for Machine Learning

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Linear Independence

Definition

A vector $\mathbf{v} \in V$ is called *linear combination* of vectors $\mathbf{x}_1, \dots, \mathbf{x}_k \in V$, if there are scalars $c_1, \dots, c_k \in \mathbb{R}$ so that

$$\mathbf{v} = c_1\mathbf{x}_1 + c_2\mathbf{x}_2 + \dots + c_k\mathbf{x}_k = \sum_{i=1}^k c_i\mathbf{x}_i.$$

Definition

The vectors $\mathbf{x}_1, \dots, \mathbf{x}_k \in V$ are called **linearly dependent**, if there is a non-trivial linear combination, such that

$$\sum_{i=1}^k c_i \mathbf{x}_i = \mathbf{0}$$

with at least one $c_i \neq 0$. If only the trivial solution exists, i.e., $c_1 = \dots = c_k = 0$, then the vectors $\mathbf{x}_1, \dots, \mathbf{x}_k$ are **linearly independent**.

When vectors are linearly dependent?

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- The vectors $\mathbf{x}_1, \dots, \mathbf{x}_k$ are linearly dependent if and only if (at least) one of them is a linear combination of the others. In particular, if one vector is a multiple of another vector, i.e., $\mathbf{x}_i = c\mathbf{x}_j, c \in \mathbb{R}$, then the set $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ is linearly dependent.

How to check linear independence?

A way of checking whether vectors $\mathbf{x}_1, \dots, \mathbf{x}_k$ are linearly independent is to write all vectors as columns of a matrix A . Gaussian elimination yields a matrix in (reduced) row echelon form

- The pivot columns indicate the vectors, which are linearly independent.
- The non-pivot columns can be expressed as linear combinations of the pivot columns on their left.

Linear Independence

Example

Are vectors $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $\begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}$ linearly dependent?

Linear Independence

Let $\mathbf{b}_1, \dots, \mathbf{b}_k$ be linearly independent vectors and

$$\mathbf{x}_j = \sum_{i=1}^k c_{ij} \mathbf{b}_i, \quad j = 1, \dots, m.$$

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Equivalently $\mathbf{x}_j = B\mathbf{c}_j$, where $B = [\mathbf{b}_1, \dots, \mathbf{b}_k]$, $\mathbf{c}_j = \begin{bmatrix} c_{1j} \\ \vdots \\ c_{kj} \end{bmatrix}$.

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Proposition

*The vectors $\mathbf{x}_1, \dots, \mathbf{x}_m$ are linearly independent **iff** the columns $\mathbf{c}_1, \dots, \mathbf{c}_m$ are linearly independent.*

Example

Consider linearly independent vectors $\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3 \in \mathbb{R}^n$ Let

$$\mathbf{x}_1 = -\mathbf{b}_1 + 2\mathbf{b}_2 + 3\mathbf{b}_3$$

$$\mathbf{x}_2 = \mathbf{b}_1 + \mathbf{b}_2 - 2\mathbf{b}_3$$

$$\mathbf{x}_3 = -\mathbf{b}_1 + 5\mathbf{b}_2 + 4\mathbf{b}_3$$

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Are the vectors $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3 \in \mathbb{R}^n$ linearly independent? Let

$$\mathbf{y}_1 = \mathbf{b}_1 - 2\mathbf{b}_2 + 3\mathbf{b}_3$$

$$\mathbf{y}_2 = \mathbf{b}_1 + \mathbf{b}_2 - 3\mathbf{b}_3$$

$$\mathbf{y}_3 = 2\mathbf{b}_1 + \mathbf{b}_2 - 3\mathbf{b}_3$$

Are the vectors $\mathbf{y}_1, \mathbf{y}_2, \mathbf{y}_3 \in \mathbb{R}^n$ linearly independent?

Basis and Rank

Definition

*Consider a set of vectors $\mathcal{A}$ in a vector space V . The set of all linear combinations of vectors in $\mathcal{A}$ is called the **span** of $\mathcal{A}$ and is denoted by $\text{span}(\mathcal{A})$, i.e.*

$$\text{span}(\mathcal{A}) = \left\{ \sum_{i=1}^k \lambda_i \mathbf{x}_i; \ k \in \mathbb{N}, \ \mathbf{x}_i \in \mathcal{A}, \ \lambda_i \in R \right\}$$

*If every vector $v \in V$ can be expressed as a linear combination of the vectors of $\mathcal{A}$, then $\mathcal{A}$ is called a **generating set** of V and we say $\mathcal{A}$ spans the vector space V .*

Definition

A **basis** $\mathcal{B}$ of a vector space V is a linearly independent subset of V that spans V . Equivalently, $\mathcal{B}$ is a basis of V if

- $\mathcal{B}$ is linearly independent
- $\mathcal{B}$ is a spanning set of V : $\text{span}(\mathcal{B}) = V$.

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Definition

A generating set $\mathcal{A}$ of V is called **minimal** if there exists no smaller set $\tilde{\mathcal{A}} \subset \mathcal{A} \subset V$ that spans V .

Proposition

The following statements are equivalent:

- *$\mathcal{B}$ is a basis of V*
- *$\mathcal{B}$ is a minimal generating (spanning) set of V*
- *$\mathcal{B}$ is a maximal linearly independent set of vectors in V , i.e., adding any other vector to this set will make it linearly dependent.*
- *Every vector $\mathbf{x} \in V$ is a unique linear combination of vectors from $\mathcal{B}$, i.e.*

$$\mathbf{x} = \sum_{i=1}^k \lambda_i \mathbf{b}_i = \sum_{i=1}^k \psi_i \mathbf{b}_i$$

and $\lambda_i, \psi_i \in \mathbb{R}, \mathbf{b}_i \in \mathcal{B}$, it follows that $\lambda_i = \psi_i, i = 1, \dots, k$.

Example

- In $\mathbb{R}^3$, the canonical/standard basis is

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

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- Different bases in $\mathbb{R}^3$ are

$$\mathcal{B}_1 = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}, \mathcal{B}_2 = \left\{ \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \right\}$$

Example

The set

$$\mathcal{A} = \left\{ \begin{bmatrix} 1 \\ 2 \\ 4 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 3 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix} \right\}$$

is linearly independent, but not a generating set (and not a basis) of $\mathbb{R}^4$:
For instance, the vector $[0, 0, 0, 1]^T$ cannot be obtained by a linear combination of elements in $\mathcal{A}$.

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How to find a basis?

A basis of a subspace $U = \text{span}[\mathbf{x}_1, \dots, \mathbf{x}_m] \subset \mathbb{R}^n$ can be found by executing the following steps:

1. Write the spanning vectors as columns of a matrix A
2. Determine the row echelon form of A ,
3. The spanning vectors associated with the pivot columns are a basis of U .

Generating Set and Basis

Example

Determine a basis for the subspace $U \subset \mathbb{R}^5$ spanned by the vectors

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \\ 0 \\ -1 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} 2 \\ 2 \\ 3 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{x}_3 = \begin{bmatrix} 1 \\ 0 \\ 2 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{x}_4 = \begin{bmatrix} 0 \\ 2 \\ -1 \\ -1 \\ -2 \end{bmatrix}.$$

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If $A \in \mathbb{R}^{m \times n}$ then $\text{rank}(A) \leq \min(m, n)$.

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Corollary

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Definition

*A matrix $A \in \mathbb{R}^{m \times n}$ has **full rank** if its rank equals the largest possible rank for a matrix of the same dimensions, i.e. $\text{rank}(A) = \min(m, n)$.
A matrix is said to be **rank deficient** if it does not have full rank.*

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- For all $A \in \mathbb{R}^{n \times n}$ holds: A is regular (invertible) if and only if A has full rank, i.e. $\text{rank}(A) = n$.

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- For all $A \in \mathbb{R}^{n \times n}$ holds: A is regular (invertible) if and only if A has full rank, i.e. $\text{rank}(A) = n$.
- For all $A \in \mathbb{R}^{m \times n}$ and all $\mathbf{b} \in \mathbb{R}^m$ it holds that the SLE $A\mathbf{x} = \mathbf{b}$ has a solution if and only if $\text{rank}(A) = \text{rank}(A|\mathbf{b})$, where $A|\mathbf{b}$ denotes the augmented matrix.

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- For $A \in \mathbb{R}^{m \times n}$ the dimension of the subspace of solutions for $A\mathbf{x} = \mathbf{0}$ is $n - \text{rank}(A)$. Later, we will call this subspace the kernel or the null space.

Example

Find the rank of the following matrices

$$A = \begin{bmatrix} 0 & 1 & 2 & 3 \\ 0 & 2 & 4 & 6 \\ 0 & 3 & 6 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 5 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 4 & 3 \\ 3 & 7 & 9 \end{bmatrix}.$$

Linear Mappings

Definition

For vector spaces V, W , a mapping $T : V \rightarrow W$ is called a linear mapping (or vector space homomorphism/ linear transformation) if

$$T(a\mathbf{x} + b\mathbf{y}) = aT(\mathbf{x}) + bT(\mathbf{y}), \quad \forall x, y \in V, a, b \in \mathbb{R}$$

or equivalently,

- $T(\mathbf{x} + \mathbf{y}) = T(\mathbf{x}) + T(\mathbf{y})$
- $T(a\mathbf{x}) = aT(\mathbf{x})$

Definition

Consider a transformation $T : V \rightarrow W$, where V, W can be arbitrary sets. Then T is called

- *injective, if for any $\mathbf{x}, \mathbf{y} \in V$ it follows that $T(\mathbf{x}) \neq T(\mathbf{y})$ if and only if $\mathbf{x} \neq \mathbf{y}$.*
- *surjective, if $T(V) = W$.*
- *bijective if it is injective and surjective.*

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- *surjective*, if $T(V) = W$.
- *bijective* if it is injective and surjective.

Definition

Let V, W be vector spaces.

- $T : V \rightarrow W$ is called **isomorphism**, if it is linear and bijective.
- $T : V \rightarrow V$ is called **endomorphism**, if T is linear.
- $T : V \rightarrow V$ is called **automorphism**, if T is linear and bijective.
- We define $id_V : V \rightarrow V$, $id_V(\mathbf{x}) = \mathbf{x}$ as the **identity mapping** in V .

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Vector spaces V and W are called isomorphic if there exists an isomorphism $T : V \rightarrow W$ and we write $V \cong W$.

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Theorem

Finite-dimensional vector spaces V and W are isomorphic if and only if $\dim(V) = \dim(W)$.

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Example

Show that $R^{m \times n} \cong R^{mn}$.

Matrix Representation of Linear Mappings

Consider a vector space V with an (unordered) basis $\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$. Let $B = (\mathbf{b}_1, \dots, \mathbf{b}_n)$ be an ordered basis, and $\mathbf{B} = [\mathbf{b}_1, \dots, \mathbf{b}_n]$ be a matrix whose columns are the vectors $\mathbf{b}_1, \dots, \mathbf{b}_n$.

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Definition

For any $\mathbf{x} \in V$ there is a unique representation

$$\mathbf{x} = \alpha_1 \mathbf{b}_1 + \dots + \alpha_n \mathbf{b}_n.$$

Then $\alpha_1, \dots, \alpha_n$ are the coordinates of $\mathbf{x}$ with respect to B , and the vector

$$[\mathbf{x}]_B = \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{bmatrix} \in \mathbb{R}^n$$

*is the **coordinate vector** of $\mathbf{x}$ with respect to the ordered basis B .*